

CXR Kafka Lab Manual

SW.13 — async events (Compose + Kafka UI)

CXR Ops Lab — `cxr-ops-lab`

2026-05-31

Abstract

SW.13: Zookeeper + Kafka broker + **Kafka UI (:8082)**. Publish CXR-shaped JSON to topic `cxr.claims.events`. Standalone lab — not wired to Claim Studio :3000 REST path. Schema: `schemas/cxr.claim-analyzed.v1.json`.

Contents

1	Architecture	2
2	Procedure	3
3	Files	4
4	UI evidence — Kafka UI (:8082)	5
5	Connection to CXR syllabus	6
6	Stop	7

Chapter 1

Architecture

Component	URL	UI?
Kafka broker	localhost:9092	CLI only
Kafka UI	http://localhost:8082	Yes
Zookeeper	(internal)	No
Claim Studio	:3000	Product UI (not connected)

Chapter 2

Procedure

```
cd cxr-ops-lab
./scripts/06-kafka-up.sh
./scripts/kafka-produce-sample.sh
```

Kafka UI → Topics → `cxr.claims.events` → Messages (Earliest).

Chapter 3

Files

File	Purpose
<code>compose.kafka.yaml</code>	Zookeeper, Kafka, Kafka UI
<code>scripts/06-kafka-up.sh</code>	Start stack
<code>scripts/kafka-produce-sample.sh</code>	Publish sample event
<code>schemas/cxr.claim-analyzed.Message</code>	Message schema
<code>evidence/SW13-kafka-verify-2026-05-27.md</code>	

Chapter 4

UI evidence — Kafka UI (:8082)

Use Kafka UI to confirm topic `cxr.claims.events` and message JSON (`claim_id`, `status`).
Screenshot optional: `evidence/SW13-kafka-ui-*.png`.

Chapter 5

Connection to CXR syllabus

Relates **M5.4** messaging design. Same JSON shape reused in SW.14 Redis cache fixture and SW.15 GraphQL `Claim` type.

Chapter 6

Stop

```
docker compose -f compose.kafka.yaml down
```